

ESTHER TRAN

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GitHub: [estherdev03](https://github.com/estherdev03) | Portfolio: esthertran.dev

PROFILE

Fourth-year Computer Science student, minor in Mathematics, at the University of Calgary and two-time Dean's Honours List recipient, seeking a Winter 2027 Software Engineering Internship. Builds and ships production systems end to end, from a hybrid-search AI assistant for Canadian immigration to a secure full-stack platform with two-factor authentication and OAuth2, each deployed and publicly accessible. Brings a strong foundation in algorithms and systems, a disciplined approach to strictly typed, testable code and the drive to learn unfamiliar technologies quickly and own work from design through deployment.

HIGHLIGHTS OF QUALIFICATIONS

- **Full-Stack Development:** Built and deployed three production web applications end to end using TypeScript, React/Next.js, NestJS, Express, and FastAPI, with frontends on Vercel and containerized backends on Railway.
- **AI & Information Retrieval:** Designed a hybrid RAG pipeline (BM25 + pgvector + Reciprocal Rank Fusion + Cohere reranking) over 516 NOC occupation groups and benchmarked four retrieval pipelines using NDCG@10.
- **Software Quality & Testing:** Built a fully deterministic, auditable rule engine covered by an automated pytest suite, uses strict typing (TypeScript, Pydantic, Zod) to catch boundary bugs before runtime.
- **Application Security:** Implemented JWT authentication, bcrypt hashing, TOTP two-factor authentication, Google/GitHub OAuth2, and an httpOnly-cookie BFF proxy, completed coursework in Network Systems Security.
- **Self-Directed Learning:** Independently studying AWS cloud infrastructure, distributed systems (Raft, CAP, sharding, Kafka/SQS), and system design patterns such as DDD and gRPC microservices.

EDUCATION

Bachelor of Science in Computer Science, Minor in Mathematics

Expected graduation: May 2027

University of Calgary, Calgary, Alberta

Relevant Courses

- **Data Structures, Algorithms & Their Analysis, Design & Analysis of Algorithms:** algorithm design, correctness proofs, and asymptotic complexity
- **Principles of Operating Systems:** processes, concurrency, scheduling, and memory management
- **Database Management Systems:** relational modelling, SQL, indexing, and transactions
- **Network Systems Security, Machine Learning:** secure system design and applied ML methods
- **Theoretical Foundations of Computer Science, Programming Paradigms:** computability, formal languages, functional and logic programming

SKILLS

- **Languages:** TypeScript, JavaScript, Python, Go, Java, SQL
- **Frontend:** React, Next.js, Tailwind CSS, Vite, Zustand, React Hook Form, Zod
- **Backend & Databases:** Node.js, NestJS, Express, FastAPI, SQLAlchemy, TypeORM, PostgreSQL, pgvector, MongoDB
- **Tools & DevOps:** Git/GitHub, Docker, Docker Compose, GitHub Actions (CI/CD), Linux shell, Postman, pytest
- **Fundamentals:** Data structures & algorithms, object-oriented design, concurrency & multithreading, distributed systems

PROJECTS

MaplePath: AI Express Entry Immigration Assistant

AI & RAG

Python 3.12, LangGraph, FastAPI, PostgreSQL/pgvector, Cohere Rerank, Next.js, Docker, pytest | [Source](#) | [Demo](#)

- Architected an event-driven LangGraph orchestrator with typed state that routes applicant intake through extraction, NOC retrieval, classification, and eligibility evaluation
- Built a hybrid search pipeline combining BM25, pgvector semantic embeddings, Reciprocal Rank Fusion, and Cohere reranking across 516 NOC 2021 unit groups

- Benchmarked four retrieval pipelines against ground-truth labelled data using hit rate and NDCG@10 to select the most accurate approach
- Engineered a 100% deterministic Pydantic rule engine for CRS points, FSW 67-point grid checks, and IELTS/CELP/PT/TEF/TCF-to-CLB conversions, verified by an automated pytest suite
- Shipped a responsive Next.js frontend with live CRS simulation and interactive NOC code exploration

Pomodoro App: Productivity & Session Tracking Platform

Backend & Security

TypeScript, NestJS 11, TypeORM, PostgreSQL, Next.js 16, Zustand, Passport.js, Docker | [Source](#) | [Demo](#)

- Developed a modular NestJS backend exposing 15 REST endpoints, with TypeORM migrations and Docker Compose for reproducible development
- Engineered authentication with bcrypt, JWT access tokens, optional TOTP two-factor authentication (QR setup and verification), and Google + GitHub OAuth2
- Implemented a session lifecycle API (start, end, current, history) managing 25-minute focus, 5-minute short break, and 15-minute long break intervals
- Built a reactive Next.js frontend with Zustand and React Hook Form + Zod, mirrored by class-validator DTOs on the server, behind an httpOnly-cookie BFF proxy
- Deployed the frontend to Vercel and the containerized backend and PostgreSQL database to Railway

MERN Canva: Browser-Based Graphic Design Tool

Full Stack & Graphics

React, Node.js, Express 5, MongoDB/Mongoose, Tailwind CSS, Cloudinary API, Passport.js | [Source](#) | [Demo](#)

- Built a multi-layer canvas editor supporting real-time drag, resize, rotate, z-index layering, font customization, and colour picking
- Architected a media pipeline connecting Express 5 to the Cloudinary API for instant uploads, asset management, and transformed image delivery
- Implemented hybrid authentication with email/password JWT login and Google OAuth 2.0 via Passport.js
- Designed six Mongoose models for designs, users, and reusable templates, and enabled high-resolution PNG/JPEG export via client-side rasterization

AWARDS & SCHOLARSHIPS

- **Dean's Honours List**, University of Calgary, Calgary, Alberta (2023–2025)
- **President's Admission Scholarship**, University of Calgary, Calgary, Alberta
- **Weeratne Memorial Scholarship**, University of Calgary, Calgary, Alberta